MAJOR THREE TYPES OF MUTATIONS

Mutation: (changes in nucleotide sequence)

maybe harmful beneficial or neutral

Type ① Point mutation

- Base Pair Substitution
  - Silent
  - Missense
  - Non-sense
- Base pair Insertion/Deletion
  - Triplet Repeats
  - Frameshift

Type ② Varaition in chromosome number

- Aneuploidy
- Hyperploidy
- Hypoploidy
- Euploidy
- Monoploidy
- Diploidy
- Polyploidy

Type ③ Variation in chromosomes structure

- Deletions
- Translocations
- Duplications
- Inversions

Somatic Mutation

① Spread in body after cell division through mitosis

② Donot pass to the offspring.

③ These occur in skin, body organs and other parts

Germline Mutation

① Passed to off-spring after production of gametes

② Occur in sperm cells and egg cells

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METHODS TO CLASSIFY MUTATION

1. Basis of Molecular nature of defect
2. Nature of phenotypic effect
3. Causative agent

WILD TYPE

Definition: common phenotype in animal population

Effect of mutation with reference to wild type

- **Forward Mutation** → Variation of wild-type phenotype to mutant phenotype
- **Reverse Mutation** → Variation of Mutant phenotype to mutant phenotype
- **Neutral Mutation** → Changing in the pattern of amino acid of protein BUT function of the protein remains un-changed

- **Conditional Mutation** → Expressed under specific condition e.g. High temp. or low temp.
- **Lethal Mutation** → Causes death of organisms in several months or years

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Loss of function: Complete/Partial loss of function after mutations
It is recessive

Gain of function: Produces new trait that is beneficial.
It is dominant

Spontaneous Mutation:
caused by natural changes in DNA

Induced Mutation:
caused by chemical, environmental and radiation changes

Mutagen:
Environmental agent that causes the mutations

Base Analog:
Chemical structure like 4 standard base pair that cause the mutation

Supressor Mutation:- (Imp. Concept):-

دو طرح کی mutation ہو گی Body میں لیکن اتنا ہوئی
کہ Mutation دوسری طرح کی Mutation کو
بھی اثر نہیں ہونے دیگا کیونکہ طرح کی
supress کر دے گی۔

That's why exhibit un-mutated wild-type.

- Different from reverse mutation
- Used for identifying new genetic sites

اصل قسم کی mutation ایک سیل میں ہو گی اور supresser mutation کسی
دوسرے سیل میں ہو گی۔

Intragenic Mutation Suppression:-

- caused as a result of suppressor mutation

دونوں طرح کی Mutation ایک ہی سیل میں ہوں گی لیکن اُن کی Gene
الگ الگ ہوں گی۔

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POINT MUTATION

1. effect single site on DNA
2. Substitution
3. Insertion/Deletion

Transition:-
(A|G) or (C|T) substitute for itself

Transversion:-
(A|G) substitutes for (C|T) and vice-versa

Types:-
- Silent Mutation
- Missense Mutation
- Non-sense Mutation
- Addition/Insertion/Deletion

Silent Mutations

1. Same Neutral type of mutation described before
2. change sequence of amino acid but the function of protein remains same.
3. No effect on the organisms phenotype

Mis-sense Mutations

1. Substitute one amino acid only but change structure of protein
2. Protein function may change or not.
3. May have a huge effect or a minimal effect.
4. Example:
Histidine → Arginine

Non-sense Mutations

1. Convert amino acid to stop codon
2. Causes protein structure to become short
3. Causes the protein to become completely non-functional
4. Sometimes only little effect.

Addition Mutation

1. Results in frame shift mutation
2. Severe effect
3. Also called as "Insertion"

Insertion Mutation

1. Addition of new base pair to amino acid sequence
2. Affect small part of gene

Deletion Mutation

1. Resulted from frame-shift mutations
2. Causes proteins non-functional

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LESSON-136
Seems Important

Non-coding Mutations: { Promoter/Enhancer Element | Trinucleotide repeats in 5 and 3' UTRs | splice sites }

Why Gene won't do Job

- AA conserved in evolution
- Deletion/Nonsense/Frame-shift
- Severe Effects
- Missense mutation (location)
- Non-conservative

Effect of Mutation on gene function

- Null allele (no activity)
- Hypomorph (low activity)
- Gain of function
- Dominant-negative (normal product)
- Neomorph (novel activity)

Mutations in Intronic Sequence

Beginning of Intron

intron کی آخری نکلی exon کی nucleotide mutation ہوتی ہے

plus sign and downstream

Middle of Intron

exon ka pehla nucleotide

minus sign position upstream

End of Intron

Middle of intron c.77+ to c.78-

plus sign

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Mutation at:

- DNA — in capitals with first affected nucleotide
- RNA — lower-case with 1st affected nucleotide
- Protein — in capitals with 1st amino acid affected

">" indicate substitution
"_" indicate range
"del" indicate deletion
"dup" → duplication
"ins" → insertion
"inv" → inversion
"c" → coding DNA
"g" → genomic sequen
"m" → mitochondrial sequence
"r" → RNA sequence
"p" → protein sequen

Mutation Nomenclature

1. Basic DNA level
2. always written with reference sequence
3. coding DNA reference sequence is preffered.
4. Good for promotor, splicing etc.
5. cDNA reference sequence used.
6. only official HGNC gene symbols used.
7. No zero nucleotide
8. nucleotide 1 is A of ATG - translation initiation codon
9. nucleotide 5' of ATG - translation initiation codon is -1, -2, -3, ——
10. Nucleotide 3' of translation stop codon is 1*, 2*, 3*, ——
11. Only + and − signs used
12. Nucleotide sequence is purely arbitrary.
13. Covers all nucleotides of a sequence.

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Genetic Disorders:-

(genetic disease as result of mutations)

- **fully penetrant:** no effect of other genes and environment
- **low-penentrance:** small role of gene with environmntal factors

may be caused due to chemicals, radiations or environment

Classification:-

1. Multifactorial disorder:-
(gene & environmental factors involved)
2. Single gene disorder:-
(only one gene involved)
3. Chromosomal disorder:-
(1000s of gene involved) (multiple organs affected)
4. Mitochondrial disorder:-
(mutation in mitochondrial DNA)
5. Somatic mutations:-
(mutation in germ cells)

Inheritance Pattern:-

1. Autosomal Dominant
(heterozygous)
2. Autosomal Recessive:-
(homozygous)
3. X-linked Recessive:-
(present on sex chromosomes)
4. Multifactorial:-
(environment & gene both effect) (common)
5. Single Gene:-
(less common)

1. Frequency of autosomal dominant disorders varies b/w 3-9.5/1000 individual.
2. Frequency of autosomal recessive disorders varies b/w 2-2.5/1000 individuals
3. Frequency of X-linked disorders 0.5-2/1000
4. Frequency of chromosomal disorders 6-9/1000
5. Frequency of multifactorial disorders 20-50/1000 individuals.

(Imp. for MCQs.)

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Autosomal Recessive Diseases:-

(homozygous) (two mutant genes involved)

∴ 1 allele of mutation from each parent → 50% chance of mutation

∴ Net chance of inheritance of mutant alleles → 25% chance of mutation

⇒ Appears in progeny of unaffected parents

⇒ Phenotypic proportions are equal

Common Diseases:-

- Bloom Syndrome
- Carpenter Syndrome
- Cystic Fibrosis
- Thalassemia
- Gaucher's Disease
- Glycogen Storage Disease
- Rotor Syndrome
- Mental retardation
- Eye Diseases

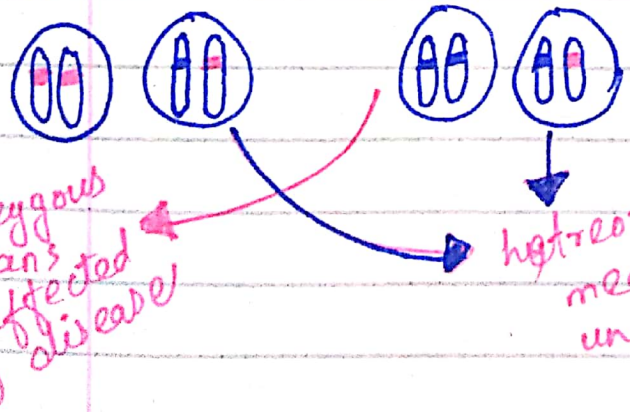
Properties:-

① Consanguinity:-
↳ common ancestors

② Identical Alleles:-
↳ same alleles from parents increase disease chance

③ Common Genetic Backgroun
↳ Small population with sam genetic background

④ Risk of disease:-
↳ 25% chances of dise in each off-spr



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Autosomal Dominant Diseases

(heterozygous)
- one-affected parent (male/female)
- 50% chance of affected off-spring

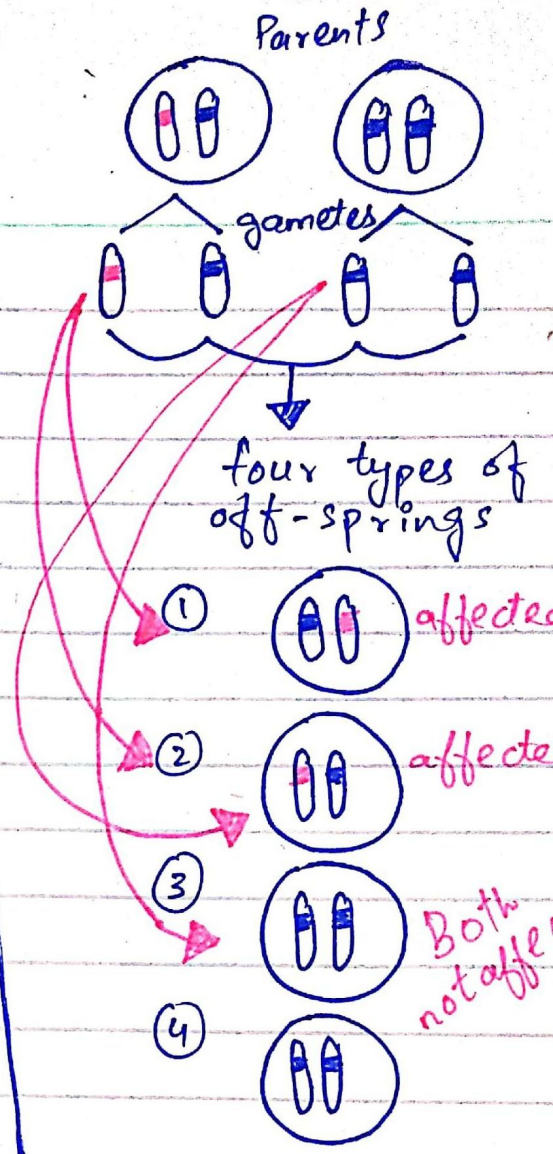
∴ Normal allele → recessive
Affected allele → dominant

Variations in autosomal dominant disease

∴ Peneterance → when genotype fail to express the phenotype

∴ Expressivity → when phenotype is severe among those with same genotype.

∴ Pleiotropy → multiple phenotypic effects on single gene



Autosomal Dominant Diseases

- Marfan Syndrome
- Huntington's disease
- Retinoblastoma
- Waardenburg syndrome
- Myotonic Dystrophy
- Polycystic Kidney Disease
- Achondroplasia
- Polydactyl
- Heredity hearing loss

Hemizygous

Males are hemizygo for loci on the X and chromosomes, where they have only a single gene.

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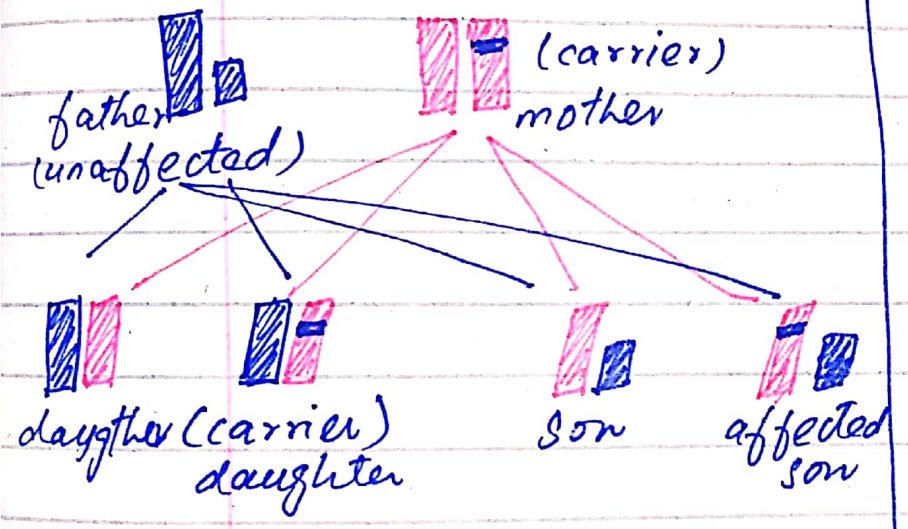
X-linked Recessive

(Present on sex chromosomes)
Mostly genes have disease.

→ Males: Hemizygous
Female: Homozygous/hetrozygous

Characteristics

* Trait is rare
* Skip Generations
* affected father don't pass to their sons
* males more affected
* only homozygous females affected.



Diseases:-

- Duchene Muscular Dystrophy
- Hunter's disease
- Menkes Disease
- Hemophilia A and B
- Color Blindness

X-Linked Dominant

Two Conditions

- affected male pass disease to all daughters but none of sons
  100% daughters, 0% sons
- affected heterozygous female and un-affected males pass disease to half sons and half daughters
  50-50% sons and daughters

Common Diseases

- Rett Syndrome
- Alport Syndrome
- Goltz Syndrome
- Fragile-X Syndrome

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Maternal Inheritance of MDNA

- inherited through orum not sperm
- mother passes disease to off-sperm
- father cannot pass disease

Characteristics

- more than one copy and one type of MDNA is passed
- mutations are common

Properties

- Heteroplasy
- Variable expression
- Pleiotropy
- Reduced Penetrance
- Functional Somatic Mosaic

Diseases

- Myoclonic epilepsy
- mitochondrial recessive ataxia
- leber hereditary optic neuropathy
- Sensory ataxia neuropathy.

Multifactorial Diseases:-

- caused by gene mutation + environmental factors

Diseases:-

- Asthma, Cancer, Diabetes
- Ciliopathies, Obesity
- Cleft palate
- Heart diseases, Hypertension
- Mood disorder, infertility
- Inflammatory Bowel disease
- intellectual disability

Daignostic Test:-

1. Physical analysis
2. Ultrasonography of Fetus
3. Analysis of aminotic fluids
4. Maternal body analysis

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DNA Polymorphism

- sequence of difference compared to a reference standard present in atleast 1-2% population
- may be due to single base or thousands bases
- May/may not effect phenotype.
- found throughout the genome.
- if location of polymorphic sequence is known, it can serve as a marker
- Each polymorphic maker has different versions (may be silent / exhibited at levels of protein)

Types of Polymorphic DNA Sequences

- RFLP: Restriction Fragment length polymorphism
- VNTR: Variable no. Tandem Repeats (8 to ≥ 50)
- STR: Short Tandem Repeats (1-8)
- SNP: Single nucleotide Polymorphism
- AFLP: Amplified Fragment length polymorphism

Why study Polymorphism?

- study inheritance pattern
- to map phenotypes.
- to anchor genes
- to determine change in function

Polymorphic DNA Sequence in Humans

- 99.9% same bases
- 0.1% different bases
- May be harmful or not

Applications of molecular markers

Short range

1) Percentage determination
2) Genetic distance estimation
3) determine twin zygosity
4) Sexing of pre-implanted embryo's
5) identification of disease carriers

long range

1) Gene mapping & QTL mapping
2) Markers assisted selection of breeds.

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Restriction Fragment Length Polymorphism:- RFLP SNP.

- Homologous DNA sequences
- Can be detected as fragments of different length after digestion of DNA samples.
- heterozygous and highly locus-specific

RFLP Probe { It is a labeled DNA sequence that hybridizes with one or more fragments of the DNA sample after they were separated by gel electrophoresis. }

Applications:-

1. Forensics application (Paternity Cases)
2. Gene Mapping
3. Analyzation of disease resistant Genes.
4. Mutation Detection
5. Determination of gene status

(e.x)

- Hungtinton Disease
- Cystic fibrosis
- Sickle cell anemia

Methodology

Genomic DNA digested by restriction enzymes

↓

DNA fragments separated by gel electrophoresis

↓

transfered to nylon membrane

↓

Membrane exposed to probes labelled with P32.

↓

Film exposed to X-Ray.

Sometimes RFLP & AFLP mein difference aa jata hai paper mein. So, that's why these DNA polymorphism topics are important :)

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VNTR: Variable Number Tandem Repeats:-

- short nucleotides sequences
- found on chromosomes
- show variation in length b/w individuals
- size of repeat is 7-10 base pairs.

→ Each Variant act as inherited allele
Rectangular Blocks represent each DNA sequence
Repeats are Tandem (Clustered in same direction)

Methodology:-

VNTP cut by restriction enzymes
↳ Blocks analyzed by PCR
↳ Amplified product run on gel electrophoresis.

Importance:-

↳ Important Source of RFLP Genetic Markers

① Removed from DNA by PCR → Size determined by gel electrophoresis → Produce unique pattern of band for each individual.

Applications

① Microbiology
② DNA fingerprinting
③ Genetic diversity
④ Forensics
⑤ Mapping of genomes
⑥ Breeding Patterns for wild and domestic animals.

Principles during VNTP data analysis

Identity Matching

↳ If two sample from same individual, then must have same allele pattern.

Inheritance Matching

↳ Every person must have a allele that matches one of each parent.

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STR: Short Tandem Repeats:-

↳ Repeats of nucleotides sequence

- Repeats region is variable b/w samples
- flanking region is constant

Homozygote: both alleles have same length

Heterozygote: both allele differ from each other

STR Markers: DNA forensics

AAAAA
(mononucleotide repeats)
ATATAT
(dinucleotide repeats)
TAGTAGTAG
(trinucleotide repeats)

TAGTTAGTTAGT
(tetranucleotide repeats)
↓
upto so on---

Applications:- {linkage analysis and DNA forensics}

Random Amplified Polymorphic DNA

- PCR based marker with 10-12 base pairs
- Random amplification
- Fragments run in agaros gel by ethidium bromide
- Unstable amplication
- Poor repeatibility

Amplified Fragment length Polymorphism

- Restriction endonucleases digestion of DNA
- ligation of adaptars
- amplification of ligated fragments
- separation via electrophoresis
- stable amplification
- good repeatibility

Molecular Markers:-

1. to study diseases
2. DNA forensics
3. QTL traits in animals and plants

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SNP: Short Nucleotide Polymorphism:-

- Every individual have distinct SNP Pattern.
- Peoples are grouped on the base of SNP
- Found in coding and non-coding regions
- Occur with high frequency 1 in 1000 to 1 in 100-300.

SNP Screening:-

- many SNPs in few individuals
- few SNPs in many individuals

Human SNPs distribution

- 25,000 (non-synonymous)
- 50,000 (synonymous)
- 25,000 (regulatory region)
- 50,000 (Intragenic)
- 50,000 (distributed intragenic)

Common Changes

Transitions

- Purine → Purine
- Pyrimidine → Pyrimidine

Transversions

- Purine → Pyrimidine
- Pyrimidine → Purine

Applications

- disease daignosis
- drug responses
- drug discovery & development
- finding predisposition to diseases
- investigation of migration pattern
- Medication and daignosis at idividual level

SNP Related Database:-

1. Androgen Receptor Mutation Database
2. Ataxia-Telangiectasia Mutation Database
3. Breast Cancer mutation Database
4. Cystic fibrosis Mutation Database
5. Cytokine Polymorphism Database
6. Fanconia Anemia mutation Database
7. GM2 Gangliosidoses Database
8. Human Ornithine Transcarbamylase Database
9. Human Type 1 & 3 Collagen Mutation Database
10. Hypertension Candidate Gene SNP Database

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mparison

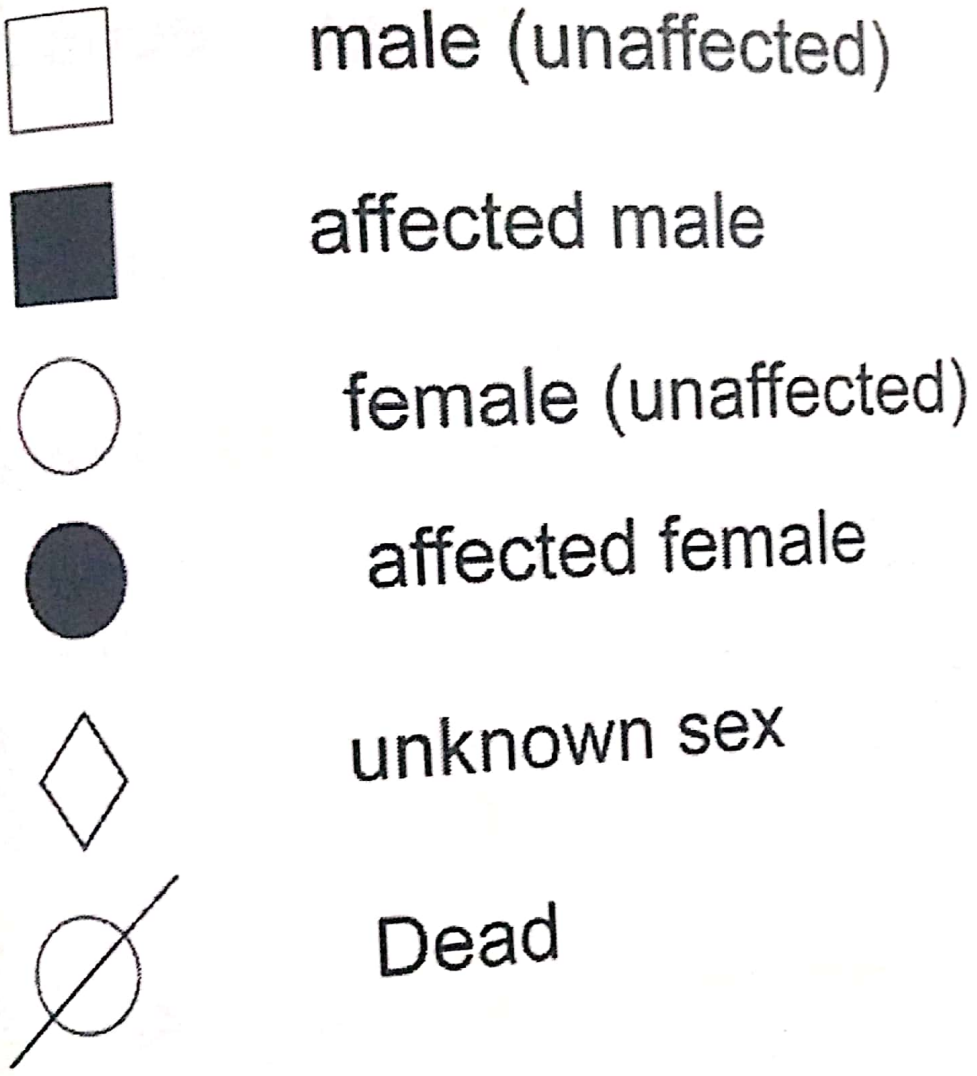
Features	RFLP	RAPD	STR	SNP
Detection method	Hybridization	PCR	PCR	PCR
Type of probe/ primer used	g DNA/ cDNA sequence of structural genes	Arbitrarily designed primer	Sequence specific primers	Sequence specific primers
Requirement of radioactivity	Yes	No/Yes	No/Yes	No/Yes
Degree of polymorphisms	Low	Medium to High	High	High

lecular Markers

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e mod sex-linked.

e probability of an affected offspring for a given cross.



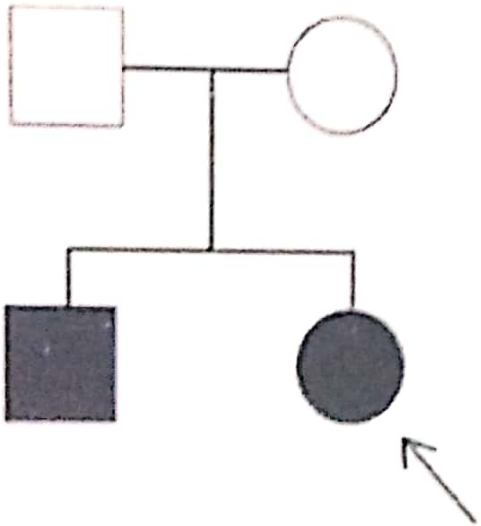
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Mating



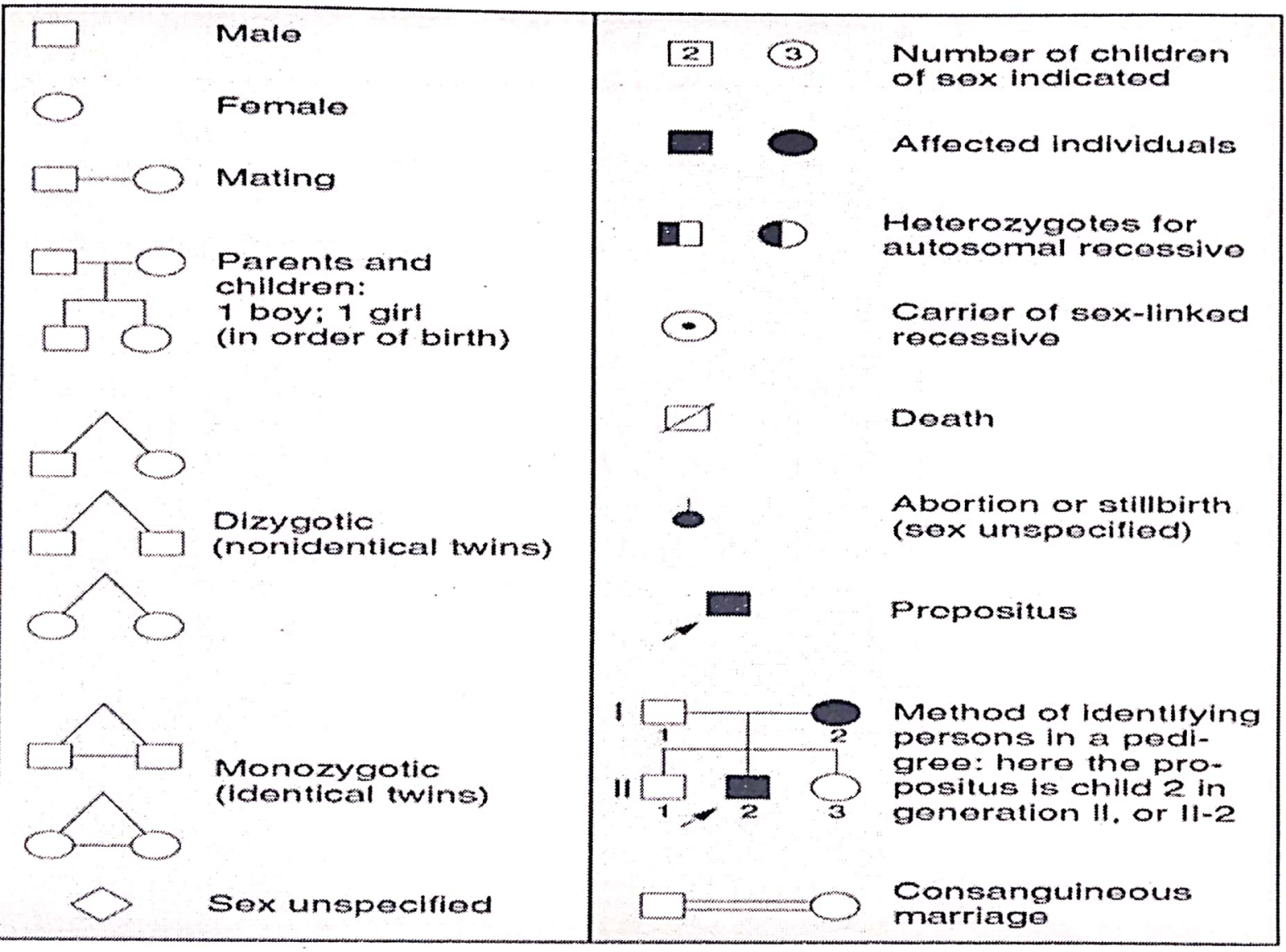
Consanguineous mating



Offspring
Arrow points to proband

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pedigree



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